Notes – Exponentially-enhanced quantum sensing with many-body phase transitions [1]

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23 June 2026 – 27 August 2026

1 Introduction

A *many-body system* is simply a quantum system that contains made up of a large number L of *interacting* particles. The interaction between them is what makes the physics interesting. Some examples we will look at in this paper are the Grover model, the p-spin model and the Biclique spin system.
To use these in sensing applications, we use these systems as a probe (many-body probe). The parameter of interest θ is made part of the system Hamiltonian. The quantum state of the L particles respond to this parameter in a sensitive way, which allows us to measure it and get a good estimate of the value of θ.
Now, if the system is on the edge of a phase transition (a change in the way the system ground state is organised), where a small change in θ will cause a transition, the sensitivity of the system will understandably increase. That is what we will try to exploit here.
Will elaborate on this more, once I get a better understanding of the methods.

2 Parameter Estimation

We'll be using the notation used in the paper, which is a bit different from what we used earlier. Recall the Symmetric Logarithmic Derivative (SLD) operator $\mathcal{L}$, implicitly defined as

$$\frac{\partial\rho(\theta)}{\partial\theta} = \frac{\rho(\theta)\mathcal{L}_\theta + \mathcal{L}_\theta\rho(\theta)}{2} \tag{1}$$

The Quantum Fisher (QFI) F^Q is then given as

$$F^Q = \mathrm{Tr}[\rho(\theta)\mathcal{L}_\theta^2] \tag{2}$$

In the case of pure states $\rho(\theta) = |\psi(\theta)\rangle\langle\psi(\theta)|$, we can obtain the form of the SLD operator, using the fact that for pure states, $\rho^2 = \rho$.

$$\begin{aligned} \partial_\theta\rho &= \partial_\theta\rho^2 \\ \Rightarrow \partial_\theta\rho &= \rho\,\partial_\theta\rho + \partial_\theta\rho\,\rho \\ \Rightarrow \frac{1}{2}(\rho\mathcal{L}_\theta + \mathcal{L}_\theta\rho) &= \rho\,\partial_\theta\rho + \partial_\theta\rho\,\rho \end{aligned} \tag{3}$$

Comparing these, we can say that for pure states,

$$\mathcal{L}_\theta = 2\partial_\theta\rho(\theta) \tag{4}$$

From results in our earlier notes, we derive the QFI for pure probe state to be

$$F^Q(\theta) = \langle\mathcal{L}_\theta^2\rangle = 4[\langle\partial_\theta\psi(\theta)|\partial_\theta\psi(\theta)\rangle - |\langle\psi(\theta)|\partial_\theta\psi(\theta)\rangle|^2] \tag{5}$$

If the Hamiltonian of the system is $H(\theta)$, then using TISE:

$$H(\theta)|\psi_n(\theta)\rangle = E_n(\theta)|\psi_n(\theta)\rangle \tag{6}$$

where $|\psi(\theta)\rangle \equiv |\psi_0(\theta)\rangle$.
Take $n = 0$ and differentiate both sides with respect to θ.

$$\begin{aligned} \partial_\theta H(\theta)|\psi_0(\theta)\rangle + H(\theta)|\partial_\theta\psi_0(\theta)\rangle &= \partial_\theta E_0(\theta)|\psi_0(\theta)\rangle + E_0(\theta)|\partial_\theta\psi_0(\theta)\rangle \\ \Rightarrow (H(\theta) - E_0(\theta))|\partial_\theta\psi_0(\theta)\rangle &= (\partial_\theta E_0(\theta) - \partial_\theta H(\theta))|\psi_0(\theta)\rangle \end{aligned} \tag{7}$$

Multiply $\langle\psi_n(\theta)|$ on both sides from the left

$$\begin{aligned}
\langle\psi_n(\theta)|(H(\theta)-E_0(\theta))|\partial_\theta\psi_0(\theta)\rangle &= \langle\psi_n(\theta)|(\partial_\theta E_0(\theta)-\partial_\theta H(\theta))|\psi_0(\theta)\rangle \\
\Rightarrow (E_n(\theta)-E_0(\theta))\langle\psi_n(\theta)|\partial_\theta\psi_0(\theta)\rangle &= \partial_\theta E_0(\theta)\langle\psi_n(\theta)|\psi_0(\theta)\rangle - \langle\psi_n(\theta)|\partial_\theta H(\theta)|\psi_0(\theta)\rangle \\
&= 0 - \langle\psi_n(\theta)|\partial_\theta H(\theta)|\psi_0(\theta)\rangle \\
\Rightarrow \langle\psi_n(\theta)|\partial_\theta\psi_0(\theta)\rangle &= -\frac{\langle\psi_n(\theta)|\partial_\theta H(\theta)|\psi_0(\theta)\rangle}{E_n(\theta)-E_0(\theta)}
\end{aligned} \tag{8}$$

Let's drop the (θ) parameter for brevity's sake. Now, we want to evaluate the first term of (5).

$$\begin{aligned}
\langle\partial_\theta\psi|\partial_\theta\psi\rangle &= \langle\partial_\theta\psi|1|\partial_\theta\psi\rangle \\
&= \sum_n \langle\partial_\theta\psi|\psi_n\rangle\langle\psi_n|\partial_\theta\psi\rangle \\
\langle\partial_\theta\psi|\partial_\theta\psi\rangle &= \sum_n |\langle\psi_n|\partial_\theta\psi\rangle|^2 \\
\Rightarrow \langle\partial_\theta\psi_0|\partial_\theta\psi_0\rangle &= |\langle\psi_0|\partial_\theta\psi_0\rangle|^2 + \sum_{n\neq 0} |\langle\psi_n|\partial_\theta\psi_0\rangle|^2
\end{aligned} \tag{9}$$

Substituting in (5), we get

$$\begin{aligned}
F^Q &= 4\left[\cancel{|\langle\psi_0|\partial_\theta\psi_0\rangle|^2} + \sum_{n\neq 0} |\langle\psi_n|\partial_\theta\psi_0\rangle|^2 - \cancel{|\langle\psi_0|\partial_\theta\psi_0\rangle|^2}\right] \\
&= 4\sum_{n\neq 0} |\langle\psi_n|\partial_\theta\psi_0\rangle|^2 \\
\therefore F^Q(\theta) &= 4\sum_{n\neq 0} \frac{|\langle\psi_n(\theta)|\partial_\theta H(\theta)|\psi_0(\theta)\rangle|^2}{(E_n(\theta)-E_0(\theta))^2}
\end{aligned} \tag{10}$$

The intuition we get from this is that if the energy gap $(E_n - E_0)$ is small, the QFI is large.

3 Bounds on the QFI

From the equation (10), we want to obtain an upper bound on the QFI. First, if we look at the denominator, it is obvious that

$$E_n - E_0 \geq E_1 - E_0 = \Delta \tag{11}$$

Thus we get the inequality

$$F^Q(\theta) \leq \frac{4}{\Delta^2}\sum_{n\neq 0} |\langle\psi_n|\partial_\theta H|\psi_0\rangle|^2 \tag{12}$$

Now to bound the numerator. We can add back the $n = 0$ term, and simplify further like so:

$$\begin{aligned}
F^Q(\theta) &\leq \frac{4}{\Delta^2}\sum_{n} |\langle\psi_n|\partial_\theta H|\psi_0\rangle|^2 \\
&= \frac{4}{\Delta^2}\sum_{n} \langle\psi_0|\partial_\theta H|\psi_n\rangle\langle\psi_n|\partial_\theta H|\psi_0\rangle \\
\Rightarrow F^Q(\theta) &\leq \frac{4}{\Delta^2}\langle\psi_0|(\partial_\theta H)^2|\psi_0\rangle \qquad \text{(completeness relation)}
\end{aligned} \tag{13}$$

For the next step, let us write $\partial_\theta H$ as A. Let A have eigenvalues a_k and eigenstates $|\phi_k\rangle$, so $A|\phi_k\rangle = a_k|\phi_k\rangle$. We can expand $|\psi_0\rangle = \sum_k c_k|\phi_k\rangle$. Thus,

$$\begin{aligned}
\langle\psi_0|A^2|\psi_0\rangle &= \sum_{j,k} c_j^* c_k \langle\phi_j|A\cdot A|\phi_k\rangle \\
&= \sum_{j,k} c_j^* c_k\, a_j^* a_k \langle\phi_j|\phi_k\rangle \\
&= \sum_k |c_k|^2 |a_k|^2 \\
&\leq \|A\|_{\text{op}}^2 \sum_k |c_k|^2 = \|A\|_{\text{op}}^2
\end{aligned} \tag{14}$$

Where $\|A\|_{\text{op}}$ is the operator norm, which, for a Hermitian operator, is equal to the maximum eigenvalue magnitude. We could stop here and say the upper bound is $\frac{4\|\partial_\theta H\|_{\text{op}}^2}{\Delta^2}$, but let us see if we can get a better one.
If we look back at (12), we see the sum $\sum_{n\neq 0}|\langle\psi_n|A|\psi_0\rangle|^2$. Let's define a new operator $A' = A - cI$, where c is some real constant (so that A' is still Hermitian). If we substitute this into the sum, we get:

$$\begin{aligned}\sum_{n\neq 0}|\langle\psi_n|A|\psi_0\rangle|^2 &= \sum_{n\neq 0}|\langle\psi_n|A'+cI|\psi_0\rangle|^2 \\ &= \sum_{n\neq 0}|\langle\psi_n|A'|\psi_0\rangle + c\langle\psi_n|\psi_0\rangle|^2 \\ \sum_{n\neq 0}|\langle\psi_n|A|\psi_0\rangle|^2 &= \sum_{n\neq 0}|\langle\psi_n|A'|\psi_0\rangle|^2 \end{aligned} \tag{15}$$

So we can rewrite (12) as

$$F^Q \leq \frac{4}{\Delta^2}\sum_{n\neq 0}|\langle\psi_n|A'|\psi_0\rangle|^2 \leq \frac{4}{\Delta^2}\|A'\|_{\text{op}}^2 \qquad \text{(from (13) and (14))} \tag{16}$$

Let us now define the value of c to be $\frac{\lambda_{\max}+\lambda_{\min}}{2}$. This shifts all the eigenvalues down, so they now range from $-\frac{\lambda_{\max}-\lambda_{\min}}{2}$ to $+\frac{\lambda_{\max}-\lambda_{\min}}{2}$. Thus, the norm of A' becomes

$$\|A'\|_{\text{op}} = \frac{\lambda_{\max}-\lambda_{\min}}{2} \tag{17}$$

Substituting this into (16), we get

$$F^Q \leq \frac{\cancel{4}}{\Delta^2}\frac{(\lambda_{\max}-\lambda_{\min})^2}{\cancel{4}} = \frac{(\lambda_{\max}-\lambda_{\min})^2}{\Delta^2} \tag{18}$$

Now instead of leaving it like this, we can define $(\lambda_{\max}-\lambda_{\min})$ to be the *seminorm* of the operator $A = \partial_\theta H$. An operator seminorm has all but one of the properties of an operator norm: It is non-negative (≥ 0), homogenous ($f(\alpha X) = |\alpha| f(X)$), and fulfils the triangle inequality. However, it does not satisfy the condition $f(X) = 0 \implies X = 0$, which the norm satisfies.
Thus, we define our operator seminorm to be the eigenvalue spread of the operator, i.e., the difference between maximum and minimum eigenvalues. Our final bound is

$$F^Q(\theta) \leq \frac{\|\partial_\theta H(\theta)\|^2}{\Delta^2} \tag{19}$$

Why did we choose c to be $\frac{\lambda_{\max}+\lambda_{\min}}{2}$? Why not something even larger, so the bound becomes even more strict? The reason for this is that we are trying to define c such that A' has the lowest possible operator norm. This means we are trying to minimise its $\max_n|\lambda_n|$. If we increase c further, then the smaller eigenvalue will have a larger magnitude, and thus the operator norm will be larger. $c = \frac{\lambda_{\max}+\lambda_{\min}}{2}$ is the optimal point.
For Hamiltonians of the form $H(\theta) = H_C + H_\theta$, where H_C is a control term, and H_θ is the θ-dependent part, then the inequality simplifies to

$$\boxed{F^Q(\theta) \leq \frac{\|\partial_\theta H_\theta\|^2}{\Delta^2}} \tag{20}$$

Here, we use the ground state $|\psi_0(\theta)\rangle$ of the system in the probe. Another way is to use a time-evolved state $|\psi(\theta,t)\rangle$ that evolves from $|\psi(\theta,0)\rangle$ with the time-dependent Hamiltonian $H(\theta,t) = H_C(t) + H_\theta$. Taking the TDSE of $|\psi\rangle$ (ignoring $\hbar$) and differentiating by θ on both sides:

$$i\frac{\partial}{\partial t}|\psi\rangle = H|\psi\rangle \tag{21}$$

$$\begin{aligned} i\partial_t|\partial_\theta\psi\rangle &= H|\partial_\theta\psi\rangle + (\partial_\theta H)|\psi\rangle \\ &= H|\partial_\theta\psi\rangle + (\partial_\theta H_\theta)|\psi\rangle \end{aligned} \tag{22}$$

Let's define $|\phi(t)\rangle = |\partial_\theta\psi(t)\rangle$. The time-derivative of its norm satisfies

$$\begin{aligned}\frac{\partial}{\partial t}\langle\phi|\phi\rangle &= \langle\partial_t\phi|\phi\rangle + \langle\phi|\partial_t\phi\rangle = 2\text{Re}[\langle\phi|\partial_t\phi\rangle] \\ &= 2\text{Re}[-i\langle\phi|H|\phi\rangle - i\langle\phi|(\partial_\theta H_\theta)|\psi\rangle] \\ &= 2\text{Re}[-i\langle\phi|(\partial_\theta H_\theta)|\psi\rangle] \end{aligned} \tag{23}$$

The final simplification is done using the fact that H is Hermitian and thus has real eigenvalues, so the $-i\langle\phi|H|\phi\rangle$ term is purely imaginary.
Taking absolute value on both sides,

$$\left|\frac{\partial}{\partial t}\langle\phi|\phi\rangle\right| = 2|\mathrm{Re}[-i\langle\phi|(\partial_\theta H_\theta)|\psi\rangle]| \leq 2|\langle\phi|(\partial_\theta H_\theta)|\psi\rangle| \tag{24}$$

We can then use Cauchy-Schwarz Inequality to say

$$\begin{aligned}\left|\frac{\partial}{\partial t}\langle\phi|\phi\rangle\right| &\leq 2\|\langle\phi|\| \cdot \|\partial_\theta H_\theta|\psi\rangle\| \\ &\leq 2\||\phi\rangle\| \cdot \|\partial_\theta H_\theta\|_{\mathrm{op}} \cdot \||\psi\rangle\| \\ \Rightarrow \left|\frac{\partial}{\partial t}\langle\phi|\phi\rangle\right| &\leq 2\||\phi\rangle\| \cdot \|\partial_\theta H_\theta\|_{\mathrm{op}}\end{aligned} \tag{25}$$

Now, we can write $\||\phi\rangle\|$ as $\sqrt{\langle\phi|\phi\rangle} = f$. Then we get

$$\begin{aligned}\left|\frac{\partial f^2}{\partial t}\right| &\leq 2f\,\|\partial_\theta H_\theta\|_{\mathrm{op}} \\ \Rightarrow \cancel{2f}\left|\frac{\partial f}{\partial t}\right| &\leq \cancel{2f}\|\partial_\theta H_\theta\|_{\mathrm{op}} \\ \Rightarrow \left|\frac{\partial f}{\partial t}\right| &\leq \|\partial_\theta H_\theta\|_{\mathrm{op}}\end{aligned} \tag{26}$$

We will now integrate on both sides, and use the fact that $f(0) = \||\phi(0)\rangle\| = 0$ (because initial state $|\psi(0)\rangle$ is prepared before θ is known).

$$\begin{aligned}\||\phi(t)\rangle\| &\leq t\,\|\partial_\theta H_\theta\|_{\mathrm{op}} \\ \Rightarrow \langle\partial_\theta\psi|\partial_\theta\psi\rangle &\leq t^2\|\partial_\theta H_\theta\|_{\mathrm{op}}^2\end{aligned} \tag{27}$$

Now, taking (5),

$$\begin{aligned}F^Q(\theta) &= 4[\langle\partial_\theta\psi|\partial_\theta\psi\rangle - |\langle\psi|\partial_\theta\psi\rangle|^2] \\ &\leq 4\langle\partial_\theta\psi|\partial_\theta\psi\rangle = 4t^2\|\partial_\theta H_\theta\|_{\mathrm{op}}^2\end{aligned} \tag{28}$$

We can use the same seminorm definition as before to get

$$\boxed{F^Q(\theta) \leq t^2\|\partial_\theta H_\theta\|^2} \tag{29}$$

There will be a $1/\hbar^2$ factor as well, which we are ignoring.

In our case, our probe state is prepared through *adiabatic processes*. The definition of "adiabatic" in quantum mechanics is not the same as it in thermodynamics. For a process to be adiabatic, it must be slow enough (i.e. the Hamiltonian must evolve slowly enough) such that if a system starts at the n-th eigenstate $|\psi_n(0)\rangle$, at any time t, it will still be in the n-th eigenstate $|\psi_n(t)\rangle$. One can see the parallels between this "adiabatic theorem" and the definition of a "quasi-static process" in classical thermodynamics.
For the process to be sufficiently slow, the total time T taken for the Hamiltonian to evolve from initial to final form should scale inversely with the energy gap Δ between the adjacent eigenstates. In our case, we prepare our probe state in the ground state, and we want it to stay in ground state throughout evolution. So T must scale as

$$T \sim \frac{1}{E_1 - E_0} = \frac{1}{\Delta} \tag{30}$$

Thus, you can see that for an adiabatic process, both (20) and (29) hold true.
These bound calculations serve the purpose of helping us realise that the QFI cannot be arbitrarily large. The properties of the system provide a maximum possible value it can have, and we will try to get as close as possible to that value.

4 Models

In the models we create to perform this method of sensing, the Hamiltonian takes the general form

$$H(\theta) = H_1 + \theta H_2 \tag{31}$$

Here, H_1 and H_2 are competing terms, each trying to pull the ground state of the system to one form. Their relative influence on the system is determined by θ. If θ is large, H_2 dominates, if θ is small, H_1 dominates.
At a certain critical value $\theta = \theta_c$, the competing terms have comparable strengths, and the system may undergo a *phase transition*, where the ground state $|GS(\theta)\rangle$ changes dramatically. When this happens, the energy gap vanishes in the *thermodynamic limit* (system size $L \to \infty$). You can see from (10) that this will mean a growth in QFI.
Now, phase transitions can be of different orders. The order is determined by the minimum n such that the nth derivative of the ground state energy (w.r.t θ) is discontinuous at θ_c. For first-order phase transitions, the energy gap decreases exponentially with system size. This is what we will exploit to get an exponential enhancement in sensing.

4.1 Grover model

4.1.1 Model Setup

This system is made up of L qubits, spanning a Hilbert space of dimension $N = 2^L$. Its Hamiltonian is

$$H_{\text{Grover}}(\theta) = \underbrace{-|m\rangle\langle m|}_{H_1} - \theta \underbrace{|\psi\rangle\langle\psi|}_{-H_2} \tag{32}$$

where $|m\rangle$ is one particular qubit configuration (L-bit binary string), and $|\psi\rangle$ is the superposition of all possible configurations.

$$|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} |j\rangle = \frac{1}{\sqrt{N}}|m\rangle + \sqrt{\frac{N-1}{N}}|m^\perp\rangle \tag{33}$$

$$\text{where} \qquad |m^\perp\rangle = \frac{1}{\sqrt{N-1}} \sum_{j \neq m} |j\rangle \tag{34}$$

What is the reason behind having minus signs in (32)*?* It is so that the two energy states $|m\rangle\langle m|$ and $|\psi\rangle\langle\psi|$ are the two lowest states of the system. We want the transition to be between the ground state and the next highest state. If the minus signs were not present, The two mentioned states would be the two highest-energy states, and the ground state would be some state orthogonal to them, with no interesting phase transition possible. The competition would be to avoid those states, rather than to go to them.
So, from (32), we have our competing terms H_1 and H_2. H_1 is pushing for the system to be completely localised on $|m\rangle$, while H_2 is pushing for the system to be completely delocalised over all configurations. In $|\psi\rangle\langle\psi|$, every configuration is equally related to every other configuration (all-to-all connectivity). These two are completely opposite configurations, which makes the phase transition more dramatic, so the sensitivity increases accordingly.
Despite it being in a 2^L-dimensional Hilbert space, we can write the Hamiltonian just in terms of $|m\rangle$ and $|m^\perp\rangle$, so it is effectively just a two-level system.

$$\begin{aligned}
H_{\text{Grover}}(\theta) &= -|m\rangle\langle m| - \theta\left[\left(\frac{1}{\sqrt{N}}|m\rangle + \sqrt{\frac{N-1}{N}}|m^\perp\rangle\right)\left(\frac{1}{\sqrt{N}}\langle m| + \sqrt{\frac{N-1}{N}}\langle m^\perp|\right)\right] \\
&= -|m\rangle\langle m| - \theta\left[\frac{1}{N}|m\rangle\langle m| + \frac{\sqrt{N-1}}{N}|m\rangle\langle m^\perp| + \frac{\sqrt{N-1}}{N}|m^\perp\rangle\langle m| + \frac{N-1}{N}|m^\perp\rangle\langle m^\perp|\right] \\
H_{\text{Grover}}(\theta) &= -\left[\left(\frac{\theta}{N}+1\right)|m\rangle\langle m| + \frac{\theta\sqrt{N-1}}{N}|m\rangle\langle m^\perp| + \frac{\theta\sqrt{N-1}}{N}|m^\perp\rangle\langle m| + \frac{\theta(N-1)}{N}|m^\perp\rangle\langle m^\perp|\right]
\end{aligned} \tag{35}$$

Thus, in matrix form:

$$H_{\text{Grover}}(\theta) = -\begin{pmatrix} \frac{\theta}{N}+1 & \frac{\theta\sqrt{N-1}}{N} \\ \frac{\theta\sqrt{N-1}}{N} & \frac{\theta(N-1)}{N} \end{pmatrix} \tag{36}$$

We can think of this like so: the system is a 2^L-dimensional room. However, this Hamiltonian H_{Grover} only moves the system along a specific 2D plane. This plane is spanned by the orthogonal vectors $|m\rangle$ and $|m^\perp\rangle$ in the space.

4.1.2 Phase Transition and Critical θ

We'll now analyse the phase transition. We start by finding the eigenvalues of this matrix. As we know the general formula of the eigenvalues of a 2×2 matrix with trace T and determinant D:

$$\lambda_{\pm} = \frac{T \pm \sqrt{T^2 - 4D}}{2} \tag{37}$$

If we calculate T and D for our Hamiltonian,

$$\begin{aligned} T &= -\left(\frac{\theta}{N} + 1 + \theta - \frac{\theta}{N}\right) \\ &= -(\theta + 1) \end{aligned} \tag{38}$$

$$\begin{aligned} D &= \frac{\theta(\theta + N)(N-1)}{N^2} - \frac{\theta^2(N-1)}{N^2} \\ &= \frac{\theta(N-1)}{N^2}(\theta + N - \theta) \\ &= \frac{\theta(N-1)}{N} \end{aligned} \tag{39}$$

Thus, the eigenvalues are

$$\begin{aligned} E_{\pm} &= \frac{-(\theta+1) \pm \sqrt{(\theta+1)^2 - 4\theta\left(1 - \frac{1}{N}\right)}}{2} \\ &= \frac{-(\theta+1) \pm \sqrt{\theta^2 + 2\theta + 1 - 4\theta + \frac{4\theta}{N}}}{2} \\ &= \frac{-(\theta+1) \pm \sqrt{(1-\theta)^2 + \frac{4\theta}{N}}}{2} \\ &= -\frac{\theta+1}{2} \pm \frac{1}{2}\sqrt{(1-\theta)^2 + \frac{4\theta}{N}} \end{aligned} \tag{40}$$

Since we only care about the energy gap between the eigenstates, we can get

$$\Delta = E_+ - E_- = \sqrt{(1-\theta)^2 + \frac{4\theta}{N}} \tag{41}$$

In order to match it with the expression in [1], we multiply numerator and denominator by $\sqrt{N}$ to get

$$\Delta = \frac{\sqrt{N^2(1-\theta)^2 + 4N\theta}}{N} \tag{42}$$

Our next aim is to find the critical value of the parameter: θ_c. It will be the value of θ that gives the minimum energy gap. So, we minimise Δ like so:

$$\begin{aligned} \frac{d\Delta}{d\theta} = \frac{1}{N}\frac{-2N^2(1-\theta) + 4N}{2\sqrt{N^2(1-\theta)^2 + 4N\theta}} &= 0 \\ \Rightarrow -2N^2(1-\theta_c) + 4N &= 0 \\ \Rightarrow N(1-\theta_c) &= 2 \\ \therefore \theta_c &= 1 - \frac{2}{N} \end{aligned} \tag{43}$$

Since $N = 2^L$ is a large number, $\theta_c \approx 1$.
An important thing to note at this point is that the phase transition is occurring between the system's eigenstates, which are not *exactly* equal to $|m\rangle$ and $|\psi\rangle$. Let us find what they actually are, to better understand what our transition is.

For simplicity, take $\delta = \Delta/2$, so that $E_\pm = -\frac{\theta+1}{2} \pm \delta$.

$$(H - E_\pm I)\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$$

$$-\begin{pmatrix} \frac{\theta}{N} + 1 + E_\pm & \frac{\theta\sqrt{N-1}}{N} \\ \frac{\theta\sqrt{N-1}}{N} & \frac{\theta(N-1)}{N} + E_\pm \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$$

$$\Rightarrow \left(\frac{\theta}{N} + 1 + E_\pm\right)v_1 + \frac{\theta\sqrt{N-1}}{N}v_2 = 0$$

$$\Rightarrow \left(\frac{\theta}{N} + 1 - \frac{\theta+1}{2} \pm \delta\right)v_1 = -\frac{\theta\sqrt{N-1}}{N}v_2$$

$$\Rightarrow \left(\frac{\theta}{N} + \frac{1-\theta}{2} \pm \delta\right)v_1 = -\frac{\theta\sqrt{N-1}}{N}v_2$$

$$\therefore \frac{v_1}{v_2} = -\frac{\frac{\theta\sqrt{N-1}}{N}}{\frac{\theta}{N} + \frac{1-\theta}{2} \pm \delta} = \frac{\frac{\theta\sqrt{N-1}}{N}}{\frac{\theta-1}{2} - \frac{\theta}{N} \mp \delta} \approx \frac{\frac{\theta\sqrt{N-1}}{N}}{\frac{\theta-1}{2} \mp \delta} \qquad \text{(for large } N\text{)} \tag{44}$$

Therefore the eigenstates can be written out (up to an overall constant) like so:

$$|E_\pm\rangle \propto \frac{\theta\sqrt{N-1}}{N}|m\rangle + \left(\frac{\theta-1}{2} \mp \delta\right)|m^\perp\rangle = \frac{\theta\sqrt{N-1}}{N}|m\rangle + \left(\frac{\theta-1}{2} \mp \frac{1}{2}\sqrt{(1-\theta)^2 + \frac{4\theta}{N}}\right)|m^\perp\rangle \tag{45}$$

Now let's see how these eigenstates behave in our two limits: $\theta \ll 1$ and $\theta \gg 1$.
1st Case: $\theta \ll 1$

- It is easier to directly look at the Hamiltonian (32). It becomes

$$H_{\text{Grover}} \approx \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}$$

- Eigenvalues are -1 and 0, and their corresponding eigenstates are $|m\rangle$ and $|m^\perp\rangle$.
- This means the ground state in this limit is $|m\rangle$.

2nd Case: $\theta \gg 1$

- In the expression for δ, we can ignore all other terms in comparison to θ^2 in this limit, so $\delta \approx \frac{\theta}{2}$.
-

$$\Rightarrow \frac{v_1}{v_2} \approx \frac{\frac{\theta\sqrt{N-1}}{N}}{\frac{\theta-1}{2} + \frac{\theta}{2}} \approx \frac{\sqrt{N-1}}{N} \qquad \text{(for } |E_-\rangle\text{)}$$

- The normalised ground state is thus

$$|E_-\rangle \approx \frac{1}{\sqrt{N}}|m\rangle + \sqrt{\frac{N-1}{N}}|m^\perp\rangle = |\psi\rangle$$

- Thus the ground state in this limit is $|\psi\rangle$.

From this, we can see the two directions θ can take us, based on its value.
3rd Case: $\theta = \theta_c \approx 1$,

- $\delta = \frac{\sqrt{0+4N}}{2N} = \frac{1}{\sqrt{N}}$
-

$$\frac{v_1}{v_2} = \frac{\frac{\sqrt{N-1}}{N}}{\mp\frac{1}{\sqrt{N}}} = \mp\sqrt{\frac{N-1}{N}} \approx \mp 1$$

- Thus, the eigenstates are $|E_\pm\rangle = \frac{1}{\sqrt{2}}(|m\rangle \mp |m^\perp\rangle)$. Both are equal superpositions of the two states.
- One can understand that this is where the energy gap between the states is at a minimum.

$$\boxed{\Delta_{\min} = \frac{2}{\sqrt{N}} = 2^{1-\frac{L}{2}}} \tag{46}$$

4.1.3 Evaluating QFI

Now we have hopefully understood the system and how it works. Let us evaluate the QFI of the sensing done using this system. We start with our earlier derived expression (10). Since this is only a two-level system, the summation only has one term.

$$F^Q(\theta) = 4\frac{|\langle E_+|\partial_\theta H|E_-\rangle|^2}{(E_+ - E_-)^2} = \frac{4}{\Delta^2}|\langle E_+|\partial_\theta H|E_-\rangle|^2 \tag{47}$$

We must calculate the derivative of the Hamiltonian.

$$\begin{aligned}\partial_\theta H &= -\frac{\partial}{\partial\theta}(|m\rangle\langle m| + \theta\,|\psi\rangle\langle\psi|) \\ &= -|\psi\rangle\langle\psi|\end{aligned} \tag{48}$$

Our next step is to evaluate $\langle E_+|\partial_\theta H|E_-\rangle$. For that, we need some representation of the eigenstates in terms of the basis for a general θ. Let us say

$$|E_-\rangle = \cos\phi|m\rangle + \sin\phi|m^\perp\rangle \qquad |E_+\rangle = -\sin\phi|m\rangle + \cos\phi|m^\perp\rangle \tag{49}$$

where ϕ is the "mixing angle" determined by the v_1/v_2 ratio we had derived earlier in (44). We are able to write the eigenstates of any 2×2 symmetric matrix in this form. From this, we get

$$\begin{aligned}\langle E_+|\partial_\theta H|E_-\rangle &= -\langle E_+|\psi\rangle\langle\psi|E_-\rangle \\ &= -\left(-\frac{\sin\phi}{\sqrt{N}} + \cos\phi\sqrt{\frac{N-1}{N}}\right)\left(\frac{\cos\phi}{\sqrt{N}} + \sin\phi\sqrt{\frac{N-1}{N}}\right) \\ &= -\frac{\sqrt{N-1}}{N}(\cos^2\phi - \sin^2\phi) - \left(\frac{N-1}{N} - \frac{1}{N}\right)\sin\phi\cos\phi \\ \langle E_+|\partial_\theta H|E_-\rangle &= -\frac{\sqrt{N-1}}{N}\cos 2\phi - \frac{N-2}{2N}\sin 2\phi\end{aligned} \tag{50}$$

Now, to relate ϕ to θ. The general formula for the mixing angle of the eigenstates of a 2×2 symmetric matrix is:

$$\tan 2\phi = \frac{2H_{12}}{H_{11} - H_{22}} \tag{51}$$

This is derived by finding the Rotation Matrix $R(\phi)$ required to diagonalise the matrix H.
We evaluate it like so:

$$\begin{aligned}\tan 2\phi &= \frac{2\frac{\theta\sqrt{N-1}}{N}}{\frac{\theta}{N} + 1 - \theta + \frac{\theta}{N}} \\ &= \frac{2\frac{\theta\sqrt{N-1}}{N}}{1 + \frac{(2-N)\theta}{N}} \\ \tan 2\phi &= \frac{2\theta\sqrt{N-1}}{N - \theta(N-2)}\end{aligned} \tag{52}$$

Let $\sqrt{[N-\theta(N-2)]^2 + 4\theta^2(N-1)} = D$. Now the sine and cosine will be

$$\sin 2\phi = \frac{2\theta\sqrt{N-1}}{D} \qquad \cos 2\phi = \frac{N-\theta(N-2)}{D} \tag{53}$$

Substituting this into (50), we get

$$\begin{aligned}\langle E_+|\partial_\theta H|E_-\rangle &= -\frac{\sqrt{N-1}}{N}\frac{N-\theta(N-2)}{D} - \frac{N-2}{2N}\frac{2\theta\sqrt{N-1}}{D} \\ &= -\frac{\sqrt{N-1}}{ND}[N - \cancel{\theta(N-2)} + \cancel{(N-2)\theta}] \\ \therefore \langle E_+|\partial_\theta H|E_-\rangle &= -\frac{\sqrt{N-1}}{D}\end{aligned} \tag{54}$$

Before substituting into QFI, let us simplify D^2:

$$\begin{aligned} D^2 &= [N-\theta(N-2)]^2 + 4\theta^2(N-1) \\ &= N^2 - 2\theta N(N-2) + \theta^2(N-2)^2 + 4\theta^2(N-1) \\ &= N^2 - 2\theta N^2 + 4\theta N + \theta^2(N^2 - \cancel{4N} + \cancel{4}) + \cancel{4N\theta^2} - \cancel{4\theta^2} \\ D^2 &= N^2(1-\theta)^2 + 4N\theta \end{aligned} \tag{55}$$

Now, substituting this into the QFI expression (47), we get

$$\begin{aligned} F^Q(\theta) &= \frac{4}{\Delta^2}\frac{N-1}{D^2} \\ &= \frac{4}{\Delta^2}\frac{N-1}{N^2(1-\theta)^2+4N\theta} \\ &= \frac{4N^2}{N^2(1-\theta)^2+4N\theta}\frac{N-1}{N^2(1-\theta)^2+4N\theta} \\ &= \frac{4N^2(N-1)}{[N^2(1-\theta)^2+4N\theta]^2} \\ \therefore F^Q(\theta) &= \frac{4(N-1)}{[N(1-\theta)^2+4\theta]^2} \end{aligned} \tag{56}$$

In the critical neighbourhood of $\theta = \theta_c = 1$, the QFI turns out to be

$$\begin{aligned} F_c^Q &= \frac{4(N-1)}{[N(0)^2+4(1)]^2} \\ &= \frac{N-1}{4} \approx \frac{2^L}{4} \end{aligned} \tag{57}$$

$$\boxed{\therefore F_c^Q = 2^{L-2}} \tag{58}$$

Comparing this to (46), we see that the exponent of QFI growth is twice that of the gap closing.
Refer Figure 1 for a visualisation of how the sharpness of the QFI around θ_c increases as system size increases.

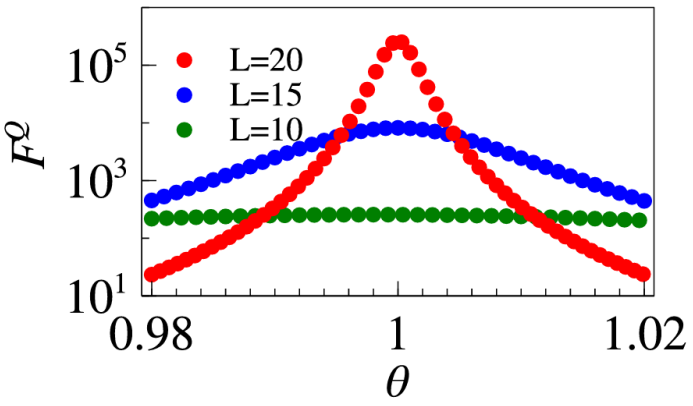


Figure 1: QFI around the critical point for different system sizes

4.1.4 Verifying the bound

We would now like to see whether the bound we derived in (20) is adhered to, and whether we get close to saturation in this model. In our case, $\|H_2\| = \|-|\psi\rangle\langle\psi|\| = 1$ (its eigenvalues are 0 and -1), so we can compare our QFI expression to the bound (20). We wish to prove that

$$F^Q(\theta) = \frac{4(N-1)}{[N(1-\theta)^2+4\theta]^2} \leq \frac{1}{\Delta^2} = \frac{N^2}{N^2(1-\theta)^2+4N\theta} \tag{59}$$

Let us cross-multiply and simplify:

$$\begin{aligned} 4(N-1)[N^2(1-\theta)^2+4N\theta] &\leq N^2[N(1-\theta)^2+4\theta]^2 \\ \Rightarrow 4(N-1)\cancel{N[N(1-\theta)^2+4\theta]} &\leq N^{\cancel{2}}[N(1-\theta)^2+4\theta]^{\cancel{2}} \\ \Rightarrow [N(1-\theta)^2+4\theta] &\geq \frac{4(N-1)}{N} \end{aligned} \tag{60}$$

We have now got the expression we want to prove. Let us start our proof with an obviously true statement: "A square is always non-negative"

$$\begin{aligned} [N(\theta-1)+2]^2 &\geq 0 \\ \Rightarrow N^2(\theta-1)^2+4+4N(\theta-1) &\geq 0 \\ \Rightarrow N^2(\theta-1)^2+4N\theta &\geq 4N-4 \\ N(1-\theta)^2+4\theta &\geq \frac{4(N-1)}{N} \end{aligned} \tag{61}$$

which is exactly what we wanted to prove in (60). Hence, we see that the bound we derived is adhered to.
Let us now see how it behaves at criticality ($\theta = \theta_c = 1$). We can see that at $\theta = 1$, the bound reduces to $N/4$. The QFI, on the other hand, reduces to $(N-1)/4$, which is equal to the bound in the thermodynamic (large N) limit.
Hence, *the bound is saturated at criticality in the thermodynamic limit.*

4.1.5 Aside: Is the Phase Transition First-Order?

While it's not particularly important, we did say at the start that all our phase transitions will be first-order here, so we should see if that is indeed the case.
A first-order phase transition is characterised by a discontinuous jump in the *order parameter*, a value that indicates how much of a process (in this case, a phase transition) is done. In this setup, the order parameter will be the expectation value of $|m\rangle\langle m|$ in the ground state. This quantifies how much of our marked state is in the ground state

$$\mathcal{O}(\theta) = \langle E_-|m\rangle\langle m|E_-\rangle = \cos^2\phi \tag{62}$$

We can understand intuitively that this order parameter will be ≈ 1 when $\theta \ll 1$, as we know the ground state is approximately $|m\rangle$, and that the parameter will be ≈ 0 when $\theta \gg 1$, as the ground state tends to $|\psi\rangle$. Our aim is to see how suddenly this transition happens (for large N). Using (53),

$$\begin{aligned} \mathcal{O}(\theta) = \cos^2\phi &= \frac{1+\cos 2\phi}{2} \\ &= \frac{1}{2}\left(1+\frac{N-\theta(N-2)}{D}\right) \end{aligned} \tag{63}$$

Now if we take the large N limit ($N \to \infty$), we first check how D reduces.

$$\begin{aligned} D &= \sqrt{N^2(1-\theta)^2+4N\theta} \qquad \text{(from (55))} \\ &= N|1-\theta|\sqrt{1+\frac{4\theta}{N(1-\theta)^2}} \\ D &\approx N|1-\theta| \end{aligned} \tag{64}$$

Substituting this into the $\cos 2\phi$ expression:

$$\begin{aligned} \cos 2\phi &\approx \frac{N-\theta(N-2)}{N|1-\theta|} \\ &= \frac{N(1-\theta)+2\theta}{N|1-\theta|} \\ &= \frac{1-\theta}{|1-\theta|}+\frac{2\theta}{N|1-\theta|} \\ \cos 2\phi &\approx \frac{1-\theta}{|1-\theta|} = \text{sgn}[1-\theta] \end{aligned} \tag{65}$$

Thus, our order parameter in the large N limit becomes:

$$\mathcal{O}(\theta) \xrightarrow{N\to\infty} \frac{1}{2}(1+\text{sgn}[1-\theta]) \tag{66}$$

$$\mathcal{O}(\theta) \xrightarrow{N\to\infty} \begin{cases} 1 & \text{for } \theta < 1 \\ 0 & \text{for } \theta > 1 \end{cases} \tag{67}$$

Thus, we see the discontinuity in the order parameter, making this a first-order phase transition.

There is still more to talk about, with regards to adiabatic state preparation and its effects on the QFI, as well as implementation concerns. We will get to these discussions after our three models are discussed, as is done in [1].

4.2 p-spin model

4.2.1 Model Setup

This model is also made up of L qubits. It will be convenient for us to think of them as "spins", that can be aligned anywhere on the Bloch sphere. The Hamiltonian is defined as follows:

$$H_{\text{p-spin}}(\theta) = \underbrace{\Bigg[-\lambda L^{1-p}\Big(\sum_{j=1}^{L}\sigma_j^z\Big)^p + (1-\lambda)L^{1-k}\Big(\sum_{j=1}^{L}\sigma_j^x\Big)^k\Bigg]}_{H_1} \underbrace{-\theta\sum_{j=1}^{L}\sigma_j^x}_{-H_2} \tag{68}$$

This is harder to understand than the Grover Hamiltonian, so we'll understand each part of it. p and k are positive integers, and $0 \le \lambda \le 1$ is an external parameter which we can use to tune the system to have either a first or a second-order phase transition (We'll discuss this more in a bit).
σ_j^z and σ_j^x are, respectively, the Pauli-Z and Pauli-X operators acting on the jth qubit.

$$\sigma_j^z = I\otimes I\otimes ...\otimes \underbrace{Z}_{j^{th}}\otimes ...\otimes I \qquad\qquad \sigma_j^x = I\otimes I\otimes ...\otimes \underbrace{X}_{j^{th}}\otimes ...\otimes I$$

Now to talk about what this Hamiltonian represents. For H_1, let us take the $\lambda = 1$ case, so that its second term vanishes. Then H_1 becomes a sum of Z-operators raised to p, while H_2 is a sum of X-operators. Thus, the competing phases differ in *two* ways:

- H_1 is pushing the system to be aligned in the up-down $|0\rangle - |1\rangle$ direction, while H_2 is pushing the system to be aligned in the left-right $|+\rangle - |-\rangle$ direction.
- H_1 shows a many-body system, where different operators interact, while H_2 shows a linear system where the qubits are independent.

These two differences push the competing terms even further apart, which makes the phase transition first-order and thus even more dramatic. If only one of these were present, the phase transition would likely be second-order and thus, much more smooth.
In the $\lambda < 1$ case, a new term is introduced, called the "antiferromagnetic fluctuation term". It is called this because it disrupts the order that would be provided by H_1 by introducing a possibility of the spin aligning in the left-right direction. Moreover, it has a positive sign unlike the other terms, so it pulls energy in the opposite direction. Thus it causes disorder much like how an antiferromagnetic term would cause disorder in a ferromagnetic system. This term also allows for us to change the phase transition from first-order to second-order.
The L^{1-p} and L^{1-k} prefactors are present to keep the system *extensive*. Energy, being extensive, should scale linearly with system size, and these prefactors make sure this proportionality is maintained.
Thus, by adjusting the parameters p, k and λ, we can not only adjust the critical point θ_c, but we can also change the nature of the phase transition.

4.2.2 Phase Transition

Let us define total magnetisation $M_z = \frac{1}{L}\sum_j \sigma_j^z$ and $M_x = \frac{1}{L}\sum_j \sigma_j^x$. Thus, the Hamiltonian can be written as

$$H_{\text{p-spin}}(\theta) = \big[-\lambda L\, M_z^p + (1-\lambda)L\, M_x^k\big] - \theta\, LM_x \tag{69}$$

Unlike the Grover model, this 2^L-dimensional Hamiltonian cannot be reduced to a 2×2 matrix, where the energy gap can then be easily calculated. It can, however, be reduced to a $(L+1)\times(L+1)$ matrix. This is due to the fact that

our Hamiltonian treats every qubit as identical, i.e., if any two qubits are swapped, the Hamiltonian is unchanged. Thus, we can use the same basis we use for spin states, which is $|S, m_s\rangle$. Here, $S = L/2$, as that is the maximum total spin number. Thus, m_s ranges from $-L/2$ to $+L/2$, which means it has $L+1$ possible values. Therefore, the Hamiltonian can be reduced to a $(L+1) \times (L+1)$ matrix.
After this reduction, the next step to find the energy gap is to diagonalise the matrix using numerical linear algebra. This can be fairly easily done just using Python. From diagonalising, we find all the eigenvalues. Thus, our energy gap is obtained.

$$\Delta(\theta, L) = E_1(\theta, L) - E_0(\theta, L) \tag{70}$$

The rest of the process is actually done numerically. Critical value θ_c is found by sweeping over θ and finding where Δ is minimum. Then we adjust p, k and λ, and test by plotting Δ_c against L to see if we obtained the exponential scaling we desire. But, for our understanding, we will try to take an analytical approach to this, with some approximations.

We can use the *Mean Field Approximation* (MFA) to get the mean energy of the system. MFA basically says that $\langle M^p \rangle \approx \langle M \rangle^p$. This holds true for large L, as the law of large numbers states that a distribution comes closer to its average as its number grows. Thus, if we defined $\langle M_z \rangle = m_z$ and $\langle M_x \rangle = m_x$,

$$\begin{aligned} E(m_z, m_x) &= L\big[-\lambda m_z^p + (1-\lambda) m_x^k - \theta m_x \big] \\ \Rightarrow \epsilon(m_z, m_x) &= -\lambda m_z^p + (1-\lambda) m_x^k - \theta m_x \end{aligned} \tag{71}$$

where ϵ is the energy "per spin".
As we are averaging over L spins, m_z and m_x (along with m_y which is not featured in the Hamiltonian), together are coordinates that describe the average spin vector. The large L approximation also allows us to ignore the discrete nature m_z and m_x actually have and treat them as continuous variables. Thus, the coordinates (m_z, m_x) is bound by the inequality $m_z^2 + m_x^2 \leq 1$, with equality only when all spins are in one direction (in the $x-z$ plane).
Now to find the minima of this energy expression. First, we can understand from looking at the signs of the energy expression (71) that all spins being aligned ("full polarisation") in the $x-z$ plane would lower the energy. Thus, we can say that the bound $m_z^2 + m_x^2 \leq 1$ is saturated. Thus, $m_z^2 + m_x^2 = 1$. We can substitute this into (71) to get

$$\epsilon(m_z) = -\lambda m_z^p + (1-\lambda)(1-m_z^2)^{k/2} - \theta\sqrt{1-m_z^2} \tag{72}$$

Now to analyse the energy landscape for first-order transition, let us take $\lambda = 1$.

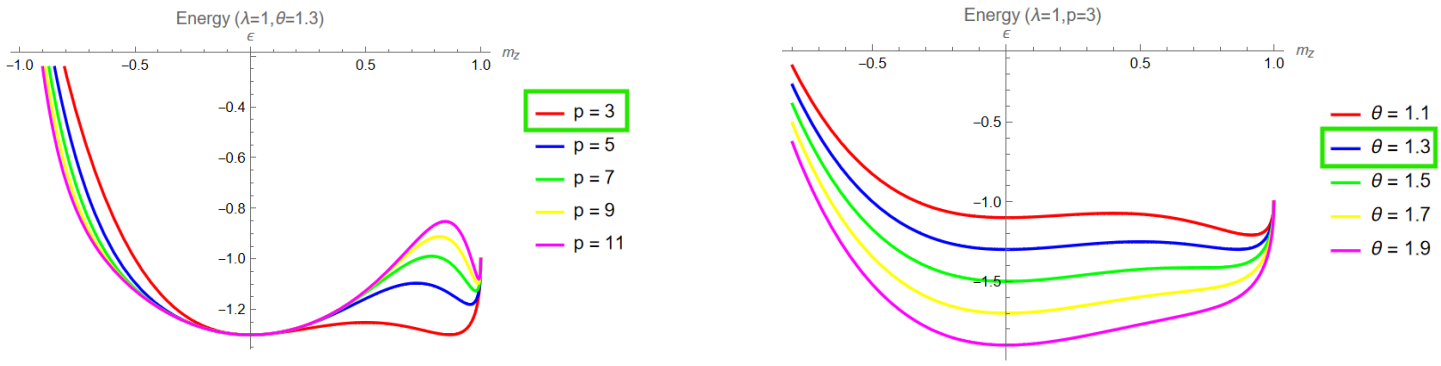
$$\epsilon(m_z) = -m_z^p - \theta\sqrt{1-m_z^2} \tag{73}$$

Let us analyse each term individually

- $\mathbf{-m_z^p}$: From analysing the shape provided by this term, we obtain two conditions on p:
  - *p must be odd:* If p is even, both $m_z = 1$ and $m_z = -1$ give the same shape, which causes a degeneracy issue, as we now have two minima (the second term is already even). This would lead to trouble finding a single defined transition point. Thus, p must be odd, so that only $m_z = 1$ would give a -1 term (minimum).
  - *p must be ≥ 3:* If $p = 1$, m_z^p would not cause any wells or interesting shapes for the energy diagram. And as we know, $p \neq 2$ as it cannot be even. Thus, $p \geq 3$.

  With this in mind, we know that $-m_z^p$ will be monotonic, with its maximum at $m_z = -1$, minimum at $m_z = +1$, and a plateau at $m_z = 0$ that gets flatter as p increases.
- $\mathbf{-\theta\sqrt{1-m_z^2}}$: This term is simpler to analyse. It is symmetric, with a minimum at $m_z = 0$.

Seeing this, we can understand that ϵ will have a minimum at $m_z = 0$ and another one near $m_z = 1$, but not exactly there. This is a *double-well configuration.* The phase transition will be the system moving from one well to the other, tunnelling through the barrier separating them. θ_c is found by calculating the θ at which the two wells are of the same depth. Refer Figure 2a to see how the location and sharpness of the second well changes with p. Refer Figure 2b to see how the relative depth of the wells changes with θ and how we find the θ_c for our fixed p. We find that $\theta_c = 1.3$ for $p = 3$. As $p \to \infty$, it turns out that $\theta_c \to 1$, which lines up with the Grover model.



(a) (b)

Figure 2: Plot of ϵ versus m_z ($\lambda = 1$) for: (a) Fixed $\theta = 1.3$, varying p, (b) Fixed $p = 3$, varying θ. The green boxes indicate the critical case, where the two wells are of equal depth, thus showing that $\theta_c = 1.3$ for $p = 3$.

Now, to understand how the energy gap closes exponentially, we use the *Wentzel–Kramers–Brillouin (WKB) approximation* for tunnelling [2]. It states that the tunnelling probability T through the barrier scales as

$$T \sim e^{-2S} \qquad S = \frac{1}{\hbar}\int_{x_1}^{x_2} \sqrt{2m(V(x) - E)}\, dx \tag{74}$$

where S is called the *Gamow factor*, m is particle mass, E is its minimum energy, $V(x)$ is the barrier potential and x_1 and x_2 are the "turning points" where $E = V(x)$.
Being probability, T is proportional to the square of the wavefunction amplitude. On the other hand, we are interested in the energy gap Δ, which is proportional to the wavefunction amplitude. Thus,

$$\Delta \sim e^{-S} \tag{75}$$

Now to obtain the expression of S in our setup. First we need to move from the extensive to the intensive case, which we do by taking a factor of L out from both the energy and the mass, giving us

$$S = \frac{L}{\hbar}\int_{x_1}^{x_2} \sqrt{2\mu_{\text{eff}}(\epsilon(x) - \epsilon_{\min})}\, dx \tag{76}$$

We can take natural units ($\hbar = 1$) for simplicity. Our variable x will be the magnetisation m_z. Thus, our expression is

$$S = L\int_{m_z^{(1)}}^{m_z^{(2)}} \sqrt{2\mu_{\text{eff}}(\epsilon(m_z) - \epsilon_{\min})}\, dm_z \tag{77}$$

Since away from θ_c, the two wells are at different energies, the value of $\epsilon_{\min}$ is ambiguous. This whole approximation only really makes sense at $\theta = \theta_c$. $m_z^{(1)}$ and $m_z^{(2)}$ are obviously the two minima points. μ_{eff} also has its own intricacies, but it is unimportant to us. All we need to understand is that $S = \alpha L$, where α is independent of L. Thus, the energy gap closes like so:

$$\Delta_c \sim e^{-\alpha L} \tag{78}$$

Thus, our energy gap closes at an exponential rate with system size. This provides an exponential growth in our QFI. Let us evaluate that QFI now.

4.2.3 Critical QFI Evaluation

We know (10).

$$F^Q(\theta) = 4\sum_{n\neq 0} \frac{|\langle\psi_n(\theta)|\partial_\theta H(\theta)|\psi_0(\theta)\rangle|^2}{(E_n(\theta) - E_0(\theta))^2}$$

Since we are not able to look at this system completely analytically, we cannot simply obtain a general QFI expression. Thus, we will only look at the critical case. In this case, $E_1 - E_0$ will be so much smaller than the remaining energy gaps, that we need not include them in our sum. Thus, the equation simplifies to

$$F_c^Q \approx 4\frac{|\langle\psi_1|\partial_\theta H|\psi_0\rangle|^2}{\Delta_c^2} \tag{79}$$

Now $\partial_\theta H = H_2 = -\sum_j \sigma_j^x = -M_x$ (from (68)). We know that it is extensive ($\propto L$). The states, being normalised, are intensive. Therefore, the numerator of the fraction is proportional to L^2. Thus,

$$F_c^Q \approx 4\frac{CL^2}{\Delta_c^2} \sim \frac{L^2}{e^{-2\alpha L}} \sim L^2 e^{2\alpha L} \tag{80}$$

Thus (ignoring the algebraic growth), in the critical case:

$$\boxed{F_c^Q \sim e^{2\alpha L}} \tag{81}$$

Therefore, we have obtained our exponentially growing QFI.

4.2.4 The $\lambda < 1$ case

We mentioned before that if $\lambda < 1$, the "antiferromagnetic fluctuation term" is introduced. This changes the system such that the phase transition is now second-order. This occurs because as θ increases, instead of the two wells reaching equal depth, they merge into a single well. Thus, no tunnelling needs to occur, and the transition happens smoothly. In Figure 3, we can see how the energy graph changes with θ when $\lambda = 0.1$, and you can see the θ_c at which the two wells merge.

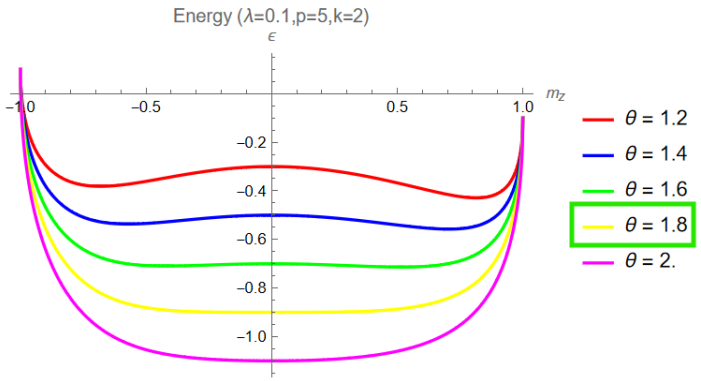


Figure 3: Plot of ϵ versus m_z ($\lambda = 0.1$, $p = 5$, $k = 2$) for varying θ. The green box indicates the critical case, where the two wells merge into one, thus showing that $\theta_c = 1.8$ for this case.

Now this case is not important for quantum sensing, as first-order phase transitions are far more useful to us. This case, however, is actually used for *Quantum Annealing* — a process by which we indirectly prepare the ground state of a complex Hamiltonian (say H_1) by first preparing the ground state of a simpler Hamiltonian (say H_2) and smoothly moving over to H_1. This is done by having the Hamiltonian $sH_1 + (1-s)H_2$ and slowly taking s from 0 to 1. This smooth transition, as you can imagine, is only possible if the phase transition is of a higher order than 1, which is why this $\lambda < 1$ case is useful.
For quantum sensing purposes, however, the $\lambda = 1$ case will suffice.
Interestingly, if we tried to perform quantum sensing using the Hamiltonian with $\lambda < 1$, our QFI would actually scale *algebraically*. The derivation of this is a bit more tedious, and as it is not particularly relevant to us, we will skip it. However, this statement provides a mathematical way for us to understand how first-order phase transitions are more useful to us than higher-order ones.

4.3 Biclique Spin System

This system is partitioned into two subsystems A and B, with L_A and L_B number of qubits respectively. These are related as $L_A = L_B + 1$, which means the total number of qubits is $L = 2L_A + 1$, which is always odd.
The Hamiltonian of this system is given below:

$$H_{\text{Biclique}} = \underbrace{\left[J\sum_{j_A=1}^{L_A}\sum_{j_B=1}^{L_B}\sigma_{j_A}^z\sigma_{j_B}^z + h_A\sum_{j_A=1}^{L_A}\sigma_{j_A}^z + h_B\sum_{j_B=1}^{L_B}\sigma_{j_B}^z\right]}_{H_1} + \theta\underbrace{\sum_{j=1}^{L}\sigma_j^x}_{H_2} \tag{82}$$

Some points:

- The first term of H_1 is the *Antiferromagnetic Ising coupling term* between subsystems A and B. J is the coupling strength ($J > 0$).
  - To minimise this term, the spins of the two subsystems must be oppositely aligned ($+1$ and -1), hence the "antiferromagnetic" label.
  - This term relates every spin in A to every spin in B, which is the definition of a "biclique" system.
- h_A and h_B are longitudinal magnetic fields, applied on A and B respectively. We say they are longitudinal since they precede a σ^z summation.
- In a similar vein, θ is a *transverse* magnetic field on the system as a whole. This is because it precedes a σ^x summation.

Here, H_1 is not pulling the system towards an all-to-all connected state, like it was in the previous two models. Here, spins within A are not coupled to each other, and the same is true for B. However, inter-subsystem coupling is complete — every spin in A is coupled to every spin in B and vice versa.

References

[1] Saubhik Sarkar, Abolfazl Bayat, Sougato Bose, and Roopayan Ghosh. Exponentially-enhanced quantum sensing with many-body phase transitions. *Nature Communications*, 16(1), June 2025. URL: `http://dx.doi.org/10.1038/s41467-025-60291-6`, `doi:10.1038/s41467-025-60291-6`.

[2] University of Manchester. WKB approximation for tunnelling. URL: `https://theory.physics.manchester.ac.uk/~judith/AQMI/PHYS30201su2.xhtml`.